

Known

$P_1 = 5 \text{ MPa}$ $T_1 = 650^\circ\text{K}$

$P_2 = 2 \text{ MPa}$ $T_3 = 640^\circ\text{K}$

$P_4 = 0.5 \text{ MPa}$

$\eta_{c1} = 90\%$ $\eta_{p1} = 90\%$

$\eta_{c2} = 100\%$ $\eta_{p2} = 100\%$

Assumptions

S.S. open (for individual components of cycle)

Closed System for Cycle

Neglect ΔPE , ΔKE

Adiabatic (for mechanical components.)

No boundary work for (B, RH, C, FWH)

Determine

y

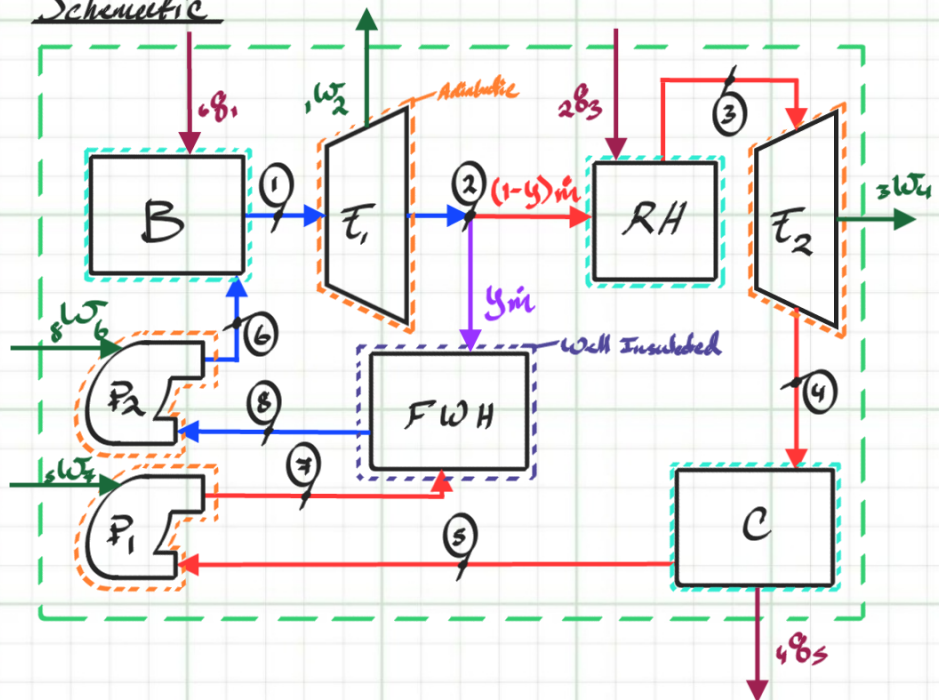
$\dot{W}_{net} = \dot{W}_{net}/\dot{m}$

$\eta_{cycle}/\dot{m}$

BWR

Draw Real state points on $t-s$ diagram for H_2O

Schematic



Symbolic

C.O.M:

$\dot{m}_1 = (1-y)\dot{m}_2 + y\dot{m}_2 = \dot{m}_2$

$\dot{m}_1 = \dot{m}_3 = \dot{m}_6 = \dot{m}$

$\dot{m}_3 = \dot{m}_4 = \dot{m}_5 = \dot{m}_7 = (1-y)\dot{m}_2$

C.O.E:

-for T_1 : $0 = \dot{Q}_1 - \dot{W}_1 + \dot{m}_1 h_1 - \dot{m}_2 h_2 \Rightarrow \dot{W}_1 = (h_1 - h_2)$

-for RH: $0 = \dot{Q}_3 - \dot{W}_3 + (1-y)\dot{m}_2 h_2 - \dot{m}_3 h_3 \Rightarrow \dot{Q}_3 = (1-y)(h_3 - h_2)$

-for T_2 : $0 = \dot{Q}_4 - \dot{W}_4 + \dot{m}_3 h_3 - \dot{m}_4 h_4 \Rightarrow \dot{W}_4 = (1-y)(h_3 - h_4)$

-for C: $0 = \dot{Q}_5 - \dot{W}_5 + \dot{m}_4 h_4 - \dot{m}_5 h_5 \Rightarrow \dot{Q}_5 = (1-y)(h_5 - h_4)$

-for P_1 : $0 = \dot{Q}_7 - \dot{W}_7 + \dot{m}_5 h_5 - \dot{m}_7 h_7 \Rightarrow \dot{W}_7 = (1-y)(h_5 - h_7)$

-for P_2 : $0 = \dot{Q}_6 - \dot{W}_6 + \dot{m}_7 h_7 - \dot{m}_6 h_6 \Rightarrow \dot{W}_6 = h_8 - h_6$

-for B: $0 = \dot{Q}_1 - \dot{W}_1 + \dot{m}_6 h_6 - \dot{m}_1 h_1 \Rightarrow \dot{Q}_1 = h_1 - h_6$

- For FWH: $\dot{Q} = \dot{Q} - \dot{W} + \sum \dot{m}_i h_i - \sum \dot{m}_e h_e$

$$\Rightarrow \dot{Q} = y \dot{m}_2 h_2 + \dot{m}_7 h_7 - \dot{m}_8 h_8$$

$$\Rightarrow \dot{Q} = y h_2 + (1-y) h_7 - h_8$$

$$\Rightarrow \dot{Q} = y(h_2 - h_7) + (h_7 - h_8)$$

$$\Rightarrow y = \frac{(h_8 - h_7)}{(h_2 - h_7)}$$

$$\dot{W}_{net}/\dot{m} = \dot{W}_{net} = \dot{W}_{in} + \dot{W}_{out}$$

$$\dot{W}_{in} = 5\dot{W}_7 + 8\dot{W}_6$$

$$\dot{W}_{out} = 1\dot{W}_2 + 3\dot{W}_4$$

$$\dot{Q}_{net}/\dot{m} = \dot{Q}_{net} = \dot{Q}_{in} + \dot{Q}_{out}$$

$$\dot{Q}_{in} = 6\dot{Q}_1 + 2\dot{Q}_3$$

$$\dot{Q}_{out} = 4\dot{Q}_5$$

$$* BWR = \left| \frac{\sum \dot{W}_p}{\sum \dot{W}_E} \right| = \left| \frac{\dot{W}_{in}}{\dot{W}_{out}} \right| = \left| \frac{(h_2 - h_6) + (1-y)(h_2 - h_7)}{(h_1 - h_2) + (1-y)(h_3 - h_4)} \right| * \eta_{cycle}/\dot{m} = \frac{\dot{W}_{net}}{\dot{Q}_{in}}$$

$$\left\{ \eta_P = \frac{\dot{W}_{net,P,S}}{\dot{W}_{net,P}} = \frac{h_i - h_{e,s}}{h_i - h_e} \quad \eta_E = \frac{\dot{W}_{net,E}}{\dot{W}_{net,E,S}} = \frac{h_i - h_e}{h_i - h_{e,s}} = \frac{T_i - T_e}{T_i - T_{e,s}} \right\} \text{ Needed Relationship for EES}$$

* Notes *

B, RH, C, FWH are isobaric $\therefore P_2 = P_3, P_4 = P_5, P_6 = P_7, P_8 = P_1$

Numerical from EES using state point properties

State	1	2	2s	3	4	4s	5	6	6s
T (°K)	650	538.3	525.9	640	463.7	463.7	425	537.1	486.1
h (kJ/kg)	3134	2433	2410	3175	2836	2836	640.1	912	
S (kJ/kg·K)	6.561	s _i =s _e		7.018	s _i =s ₄	s _i =s ₄	1.86		

State	7	7s	8
T (°K)		425.1	485.5
h (kJ/kg)	641.9	641.7	908.5
S (kJ/kg·K)	2.447		

$$\begin{aligned}
 {}_1w_2 &= 205.7 \text{ kJ/kg} & {}_2q_3 &= 213.8 \text{ kJ/kg} & w_{in} &= -5.155 \text{ kJ/kg} \\
 {}_3w_4 &= 300.1 \text{ kJ/kg} & {}_4q_5 &= -1940 \text{ kJ/kg} & w_{out} &= 505.9 \text{ kJ/kg} \\
 {}_5w_6 &= -1.608 \text{ kJ/kg} & {}_6q_1 &= 2227 \text{ kJ/kg} & q_{in} &= 2441 \text{ kJ/kg} \\
 {}_8w_6 &= -3.527 \text{ kJ/kg} & & & q_{out} &= -1940 \text{ kJ/kg}
 \end{aligned}$$

$$w_{net} = 500.7 \text{ kJ/kg}$$

$$q_{net} = 500.7 \text{ kJ/kg}$$

$$\eta_{cyc} = 20.52\%$$

$$BWR = 10.15\%$$

$$y = 0.1163$$

Diagram

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